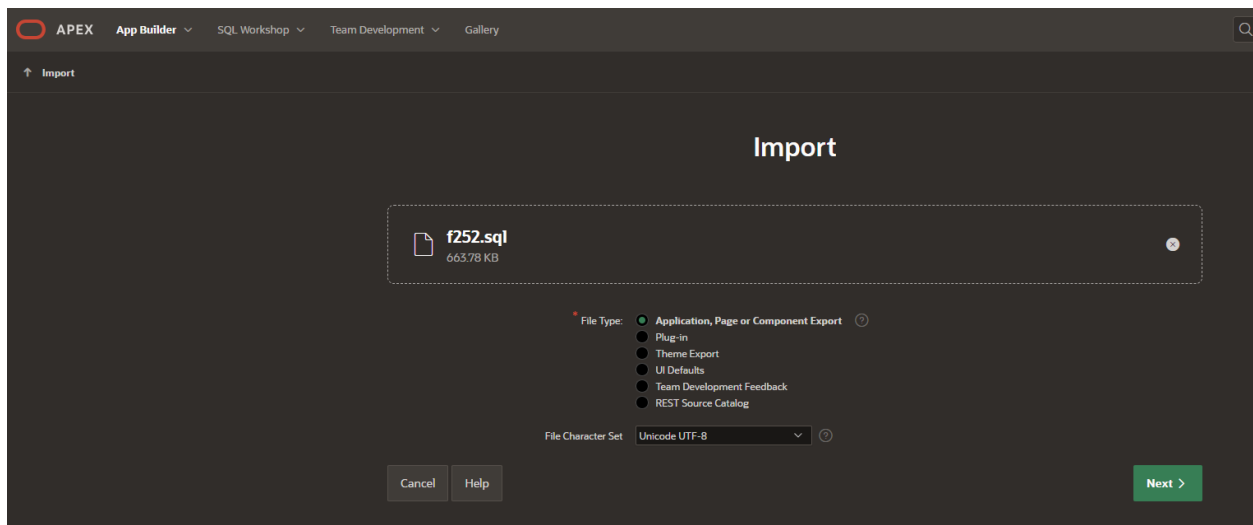


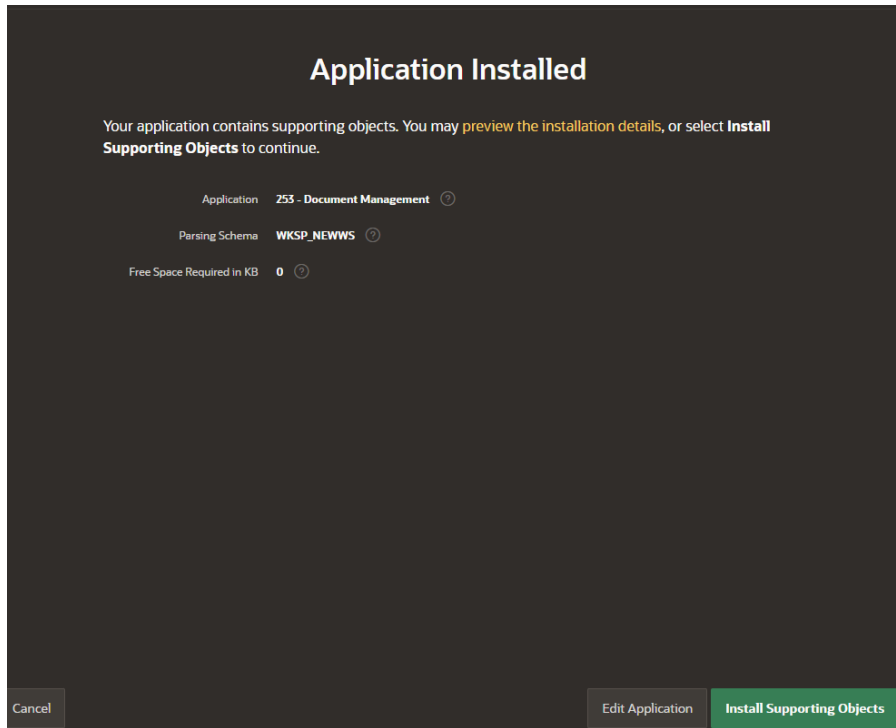
Document Management App – Installation Guide

Step 1. Application Installation

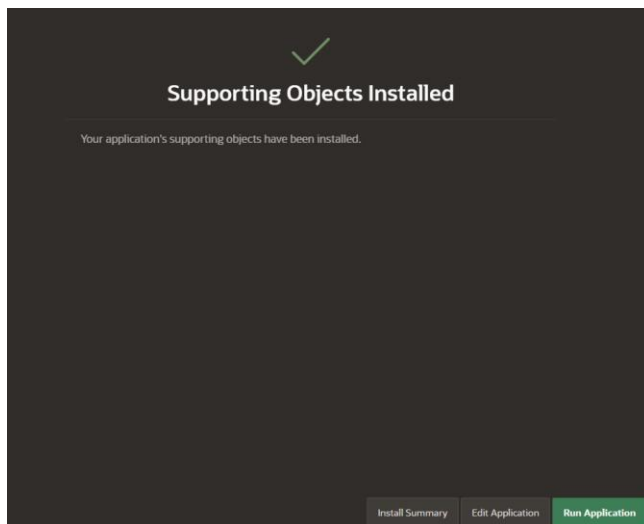
1. Create a workspace or use an existing one.
2. Login to APEX workspace.
3. Go to App Builder -> Import. Drag & Drop the app file and click on **Next** and follow the prompts to upload the app.



4. APEX will prompt to **Install Supporting Objects**. You can review the Install Summary. Once the objects are installed, a success message will appear.



5. Click **Run Application** to launch the app.



Step 2 Additional Configuration

1. This application is using Vector Search, a feature of Oracle Database 26ai. You can import and ONNX model into the Oracle Database and generate embeddings. You need to login as an administrator and grant the necessary privileges to the parsing schema: create mining model and execute on dbms_vector and dbms_cloud packages.

```
grant create mining model to YourSchemaName;
grant execute on dbms_vector to YourSchemaName;
grant execute on dbms_cloud to YourSchemaName;
```

2. Next step is to load the ONNX model into the database. Refer to the [documentation](#) to download the model (all-MiniLM-L12-v2), unzip and load it. Make sure to give “**DOC_MODEL**” as model name.

Example of loading the model stored in an Object Storage bucket, using a pre-authenticated request:

```
DECLARE
  L_PAR_URL  VARCHAR2(1000);
  L_RESPONSE_BLOB BLOB;
BEGIN
  L_PAR_URL := 'INSERT_HERE_THE_LINK_TO_MODEL'
  ;
  L_RESPONSE_BLOB := APEX_WEB_SERVICE.MAKE_REST_REQUEST_B(
    P_URL      => L_PAR_URL,
    P_HTTP_METHOD => 'GET'
  );
  DBMS_VECTOR.LOAD_ONNX_MODEL(
    MODEL_NAME => 'DOC_MODEL',
    MODEL_DATA => L_RESPONSE_BLOB,
    METADATA =>
      JSON(
        '{
          "function": "embedding",
          "embeddingOutput": "embedding",
          "input":{"input": ["DATA"]}'
        )
      );
END;
```

Example of loading the model stored in an Object Storage bucket, using credentials:

■ *Create the credentials*

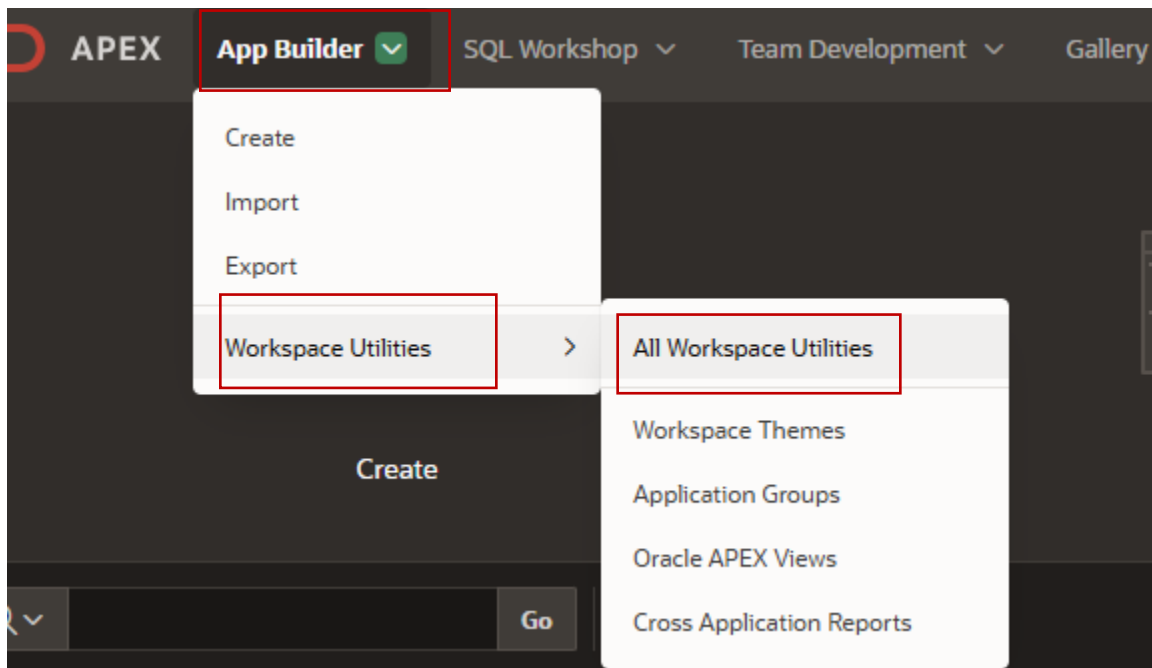
```
BEGIN
  dbms_cloud.create_credential (
    credential_name => 'onnx_obj_store_cred',
    username       => '<Your username>',
    password       => '<AUTH Token>'
  );
END;
```

■ *Load the model*

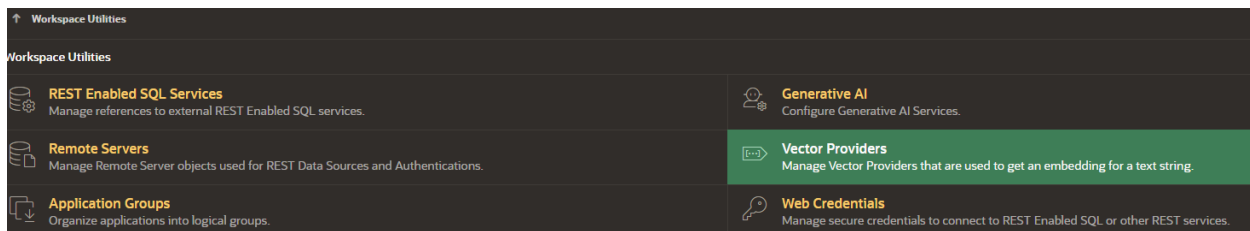
BEGIN

```
DBMS_VECTOR.LOAD_ONNX_MODEL(  
  model_name => 'DOC_MODEL',  
  model_data => dbms_cloud.get_object (  
    credential_name => 'obj_store_cred', -- replace with your credential name  
    object_uri    => '<Enter Your Object Storage URI>', -- blob  
    metadata => JSON('{  
      "function" : "embedding",  
      "embeddingOutput" : "embedding",  
      "input":{"input": ["DATA"]}  
    }')  
  );  
END;
```

3. Modify the vector provider that was imported to point to your parsing schema. Learn more about vector providers from [this](#) blog. Click on **App Builder** -> **Workspace Utilities** -> **All Workspace Utilities**.



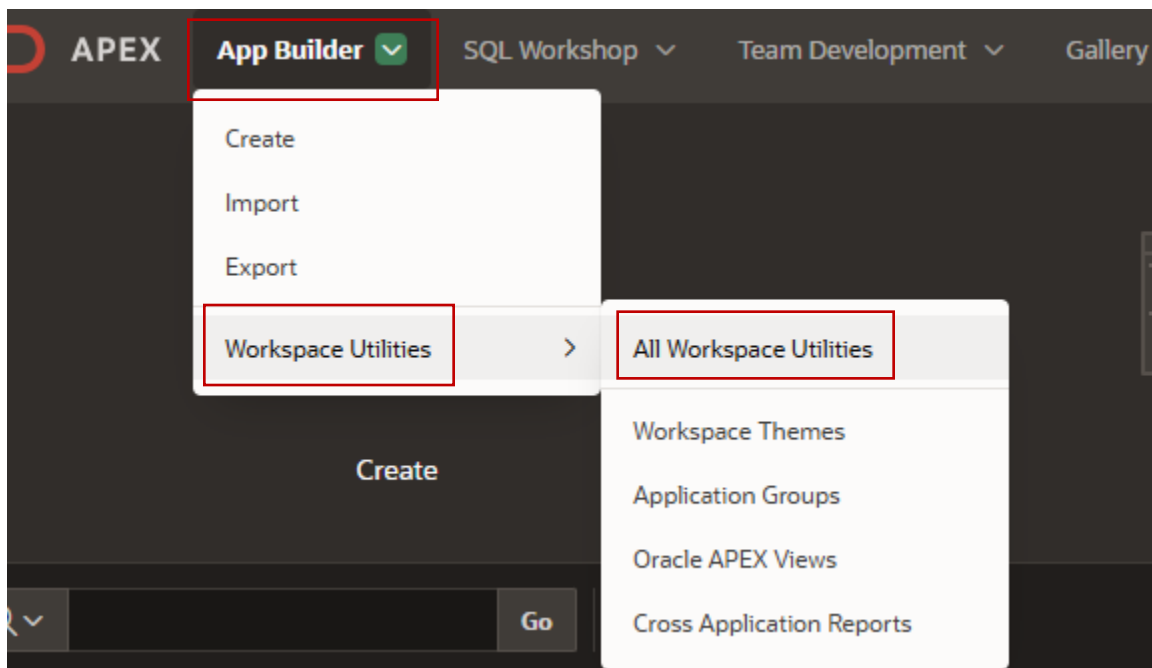
Click on **Vector Providers** and select **db_onnx_model**.



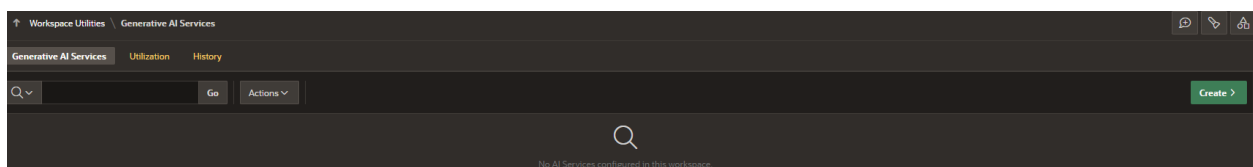
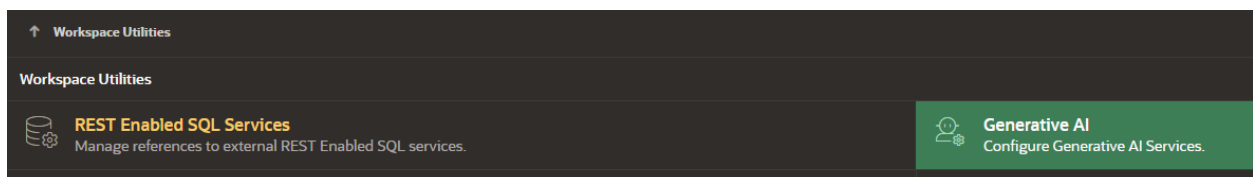
Workspace Utilities \ Vector Providers			
Vector Providers History			
Go Actions			
Name	Static ID	AI Provider	Provider Type
db_onnx_model	db_onnx_model	Local	Database ONNX Model

Modify the **ONNX Model Owner** to point to your schema and select the **ONNX Model Name** imported earlier. Click on **Apply Changes**.

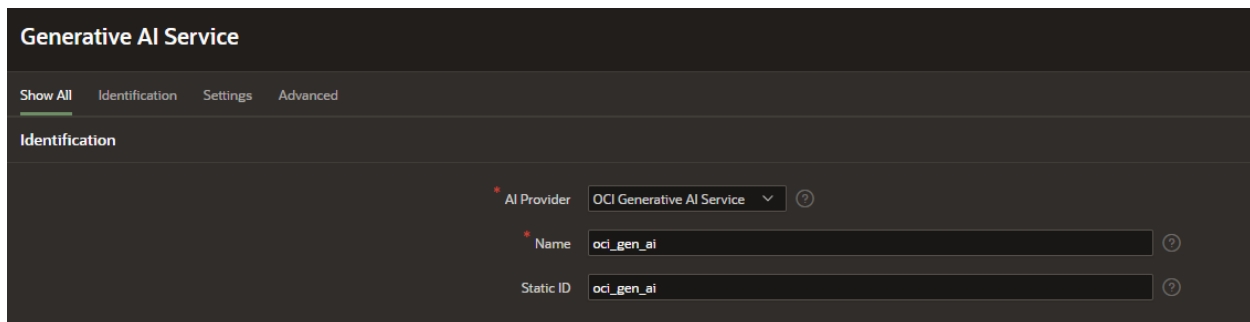
- The application leverages APEX AI capabilities for end users. It is using the two AI dynamic actions: Generate Text with AI and Show AI Assistant. To use these capabilities, you need to configure Generative AI Services at the workspace level. You have three options: OCI Generative AI, Open AI and Cohere. Click on **App Builder** -> **Workspace Utilities** -> **All Workspace Utilities**.



Click on **Generative AI** and then on **Create** button.



Select your AI Provider and enter the required details. Refer to the [documentation](#) for additional information. Type “oci_gen_ai” for name and static id.



Generative AI Service

Show All Identification Settings Advanced

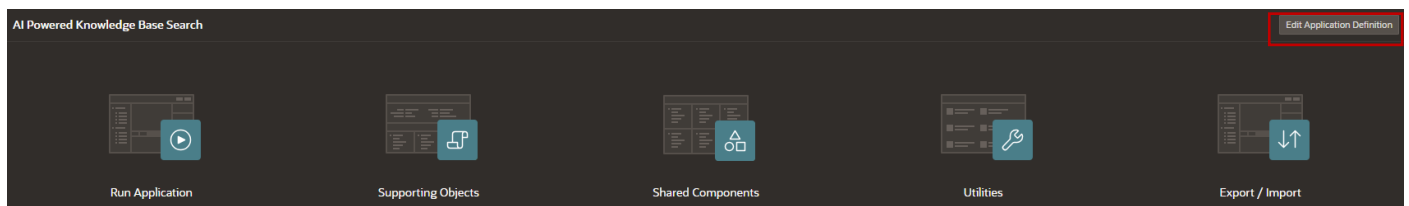
Identification

AI Provider OCI Generative AI Service ?

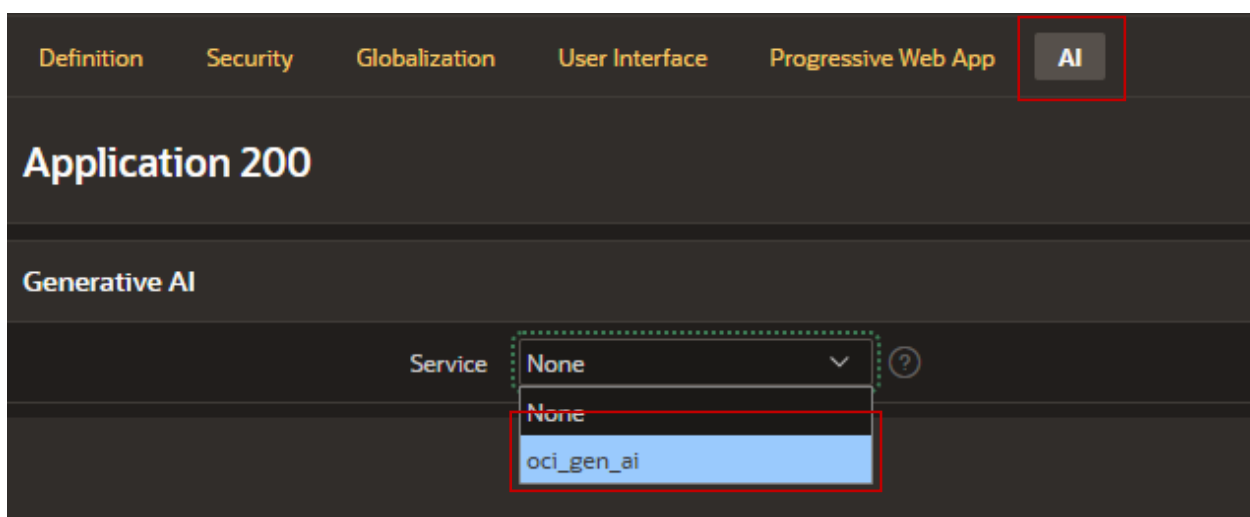
Name oci_gen_ai ?

Static ID oci_gen_ai ?

Edit the application definition to have this Generative AI Service as default service. Open the application in App Builder. Click on **Edit Application Definition**.



Click on **AI** tab and select the oci_gen_ai service. Then, click on **Apply Changes**.



Definition Security Globalization User Interface Progressive Web App AI

Application 200

Generative AI

Service None ?

None

oci_gen_ai

- Depending on the document type, different approval levels may be required. The DOCUMENT_TYPE table defines each document type and its corresponding number of approval levels, while the EMPLOYEE table stores the team or organizational hierarchy. For example, if a document is classified as Confidential, the required approval level is 2. This means the document must be approved by both the employee’s direct manager and the next-level manager.

DOCUMENT_TYPE					
Columns Data Indexes Constraints Grants Statistics Triggers Dependencies DDL Sample Queries					
+ Insert Row Columns... Filter... Count Rows Load Data Download Refresh					
	ID	TYPE	DESCRIPTION	APPROVAL_LEVEL	
	2	Private	This type of document requir...	1	
	3	Confidential	This type of document requir...	2	
	1	Public	This type of document does...	0	
	4	Highly-Confidential	This type of document requir...	3	

EMPLOYEES									
Columns Data Indexes Constraints Grants Statistics Triggers Dependencies DDL Sample Queries									
+ Insert Row Columns... Filter... Count Rows Load Data Download Refresh									
	EMPLOYEE_ID	FIRST_NAME	LAST_NAME	EMAIL	PHONE	EMPLOYEE_TYPE..	DEPARTMENT_ID..	ROLE	MANAGER_ID
	2	AMY		amy@example...		Full Time	2	Data Developer	9
	6	Peter		peter@exampl...		Full Time	1	HR Manager	7
	7	KING		king@example...		Full Time	1	General Manager	
	9	Mike		mike@exampl...			2	IT Manager	6

You can create the users **AMY**, **MIKE**, **PETER**, and **KING**, then navigate to **App XXX → Shared Components → Application Access Control**. From there, add each user and assign them the appropriate role, such as Contributor or Administrator.

Application 254 | Shared Components | Application Access Control

Application Access Control

Show All Roles User Role Assignments

Roles

Role	Static Identifier	Users	Description
Administrator	ADMINISTRATOR	0	Role ass...
Contributor	CONTRIBUTOR	0	Role ass...
Reader	READER	0	Role ass...

User Role Assignments

User Name: AMY, MIKE, PETER

Contributor

User Assignment

About Application Users

Application users are not exported as part of your application. When you deploy your application you will need to manually manage your user to role assignments. Roles are exported as part of an application export and imported with application imports.

User Name: KING

Application Role: ☐ Administrator ☒ Contributor ☐ Reader

Cancel Create Assignment

Add Role > Add User Role Assignment >

That's it! You are ready to run and explore the application. Upload PDF documents and ask questions!